



Astronaut Sunita Williams enjoys chicken and rice aboard the ISS

GROWING PLANTS

Long missions beyond Earth orbit will need to be self-sufficient, so space agencies are carrying out experiments to understand how exposure to space conditions affects plants and their seeds. Crops grown in water and nutrients (hydroponics) or in the soil of other planets could one day provide food and help to refresh air supplies.



Seedlings can grow well even with very little gravity in space.

Astronaut T. J. Creamer tends tree seedlings on the ISS

EATING DINNER

Food arrives at the ISS on unmanned cargo spacecraft, usually in sealed packs that can be rehydrated from water valves in the station's service module. Ovens reheat canned or packet food, but their temperature is limited for safety reasons.

LIFE ON MARS

Future explorers on Mars will face unique challenges, and engineers are already testing solutions on Earth. Astronauts will need lightweight, flexible spacesuits to work in Martian gravity. They will also have to be protected from dangerous radiation, perhaps by living and working mostly in underground habitats.

FAST FACTS

- The changing alignments of the planets as they orbit the sun mean that crewed missions to Mars would have to wait for about three years before Earth was close enough for them to return home.
- Russian cosmonauts set an unbroken series of records for long-duration spaceflight aboard the Mir space station in the 1990s.

▲ SIMULATING LIFE ON MARS

American researchers in the Utah desert wear prototype spacesuits and practice procedures for a future exploration mission on the surface of Mars.

